

Third Comment of the Obsidian Watch Group

Docket: HHS-OASH-2026-0232

Submitted by: Jay Puckett, Principal Security Researcher, Obsidian Watch Group

SUMMARY

The Request for Information asks how a threshold "should apply across botanical source material, semi-synthetic derivatives, and finished dosage forms." This comment answers that question in one specific and, we believe, unaddressed respect.

Every substance covered by the Department's recommendation and by DEA's orders sits on a single chemical scaffold. A second scaffold in the same plant, carrying a mu-opioid partial agonist, is not covered by anything. At least one state has already found it necessary to cover it. We ask the Department to say whether that was considered.

1. WHAT IS COVERED

DEA proposed rule, 6 July 2026 (DEA-2026-1190):
7-hydroxymitragynine above a specified threshold.

DEA temporary order, effective 26 August 2026 (91 FR, doc. 2026-17429):
mitragynine pseudoindoxyl;
MGM-15, also known as dihydro-7-hydroxymitragynine;
MGM-16, the 9-fluoro analogue.

The Department's letter of 6 March 2026 to Administrator Cole:
7-hydroxymitragynine only, by percentage and by an absolute limit of
1.00 milligram per article.

All four substances are structural relatives of mitragynine. Every one is on the mitragynine scaffold.

2. WHAT IS NOT COVERED

9-hydroxycorynantheidine is a minor indole alkaloid of *Mitragyna speciosa*, comprising less than one percent of total alkaloid content in the plant. It is a partial agonist at the mu-opioid receptor. It stands to corynantheidine approximately as 7-hydroxymitragynine stands to mitragynine: a hydroxylated minor alkaloid with mu-opioid activity, present naturally in quantities too small to extract economically.

It is not named in the proposed rule, in the temporary order, in the Department's letter of 6 March 2026, or in the FDA scientific assessment supporting the threshold, except as words appearing inside the title of a cited journal article.

We searched the federal docket system for "9-hydroxycorynantheidine" and found no documents. We searched Senate lobbying disclosures and found no filings. We searched FDA's adverse event reporting system and found no reports. Each search was run against a control that returns results.

3. A STATE HAS ALREADY COVERED IT

Florida's emergency rule of June 2026 names 9-hydroxycorynantheidine among seven covered compounds.

We note only the fact. A state regulator identified this compound and wrote it into a rule, and the federal instruments issued in the following weeks did not.

4. THE DEPARTMENT'S OWN REASONING REACHES IT

In scheduling MGM-16, DEA stated:

"While no evidence supports the presence of MGM-16 on the *Mitragyna speciosa* consumer market, its profile as a highly potent opioid that is structurally related to 7-hydroxymitragynine lends itself as an [alternative]..."

That is a preemptive scheduling of a substance not on the market, on the express reasoning that sellers will substitute toward structurally related opioids.

We agree with that reasoning. We observe that it was applied within one chemical family and not extended to the other.

If the concern is substitution, the relevant question is not whether a compound resembles 7-hydroxymitragynine specifically, but whether it is an accessible mu-opioid agonist obtainable from the same plant, through the same supply chain, sold through the same

retail channel, under the same botanical framing. On that test, 9-hydroxycorynantheidine qualifies and MGM-16, which is not on the market, also qualified.

5. WHY THIS MATTERS ON THIS DOCKET AND NOT LATER

Once the threshold takes effect, the commercial incentive to move to an unscheduled mu-opioid alkaloid from the same plant is immediate and obvious. That transition, if it occurs, will not be visible in adverse event data for many months, because reporting lags exposure.

The record before the Department is presently empty on this compound. That is an argument for addressing it now, while a rulemaking record is open, rather than after a consumer harm signal appears.

6. THE CAPABILITY TO MANUFACTURE IT ALREADY EXISTS

Two patent families claim industrial production of this compound. Both have priority dates in late 2023.

Mitradyne Corporation, Toronto, Ontario

EP 4 538 367 A2, application 24201792.9, published 16 April 2025

Priority 20 September 2023, from US provisional 63/583,976

"Polypeptides for use in the synthesis of kratom alkaloids"

- Claim 5. A polypeptide encoding a hydroxylase that hydroxylates the indole ring of corynantheidine at the C9 position to produce 9-hydroxycorynantheidine.
- Claim 6. A polypeptide encoding a hydroxylase that hydroxylates the indole ring of mitragynine at the C7 position to produce 7-hydroxymitragynine.
- Claim 8. A polypeptide encoding an O-methyltransferase that methylates the hydroxyl group at the C9 position of 9-hydroxycorynantheidine to produce mitragynine.
- Claim 11. The system of claim 10, wherein the system is cell-free.

Claims 5 and 6 are consecutive. The same document claims the enzyme that makes the compound the Department is scheduling and the enzyme that makes the compound it is not.

CB Therapeutics, Inc., Carlsbad, California
US 2025/0179544 A1, application 18/965,429
Priority 1 December 2023

"Production of corynanthe-type monoterpene indole alkaloid compounds"

This second filing claims recombinant-cell production, that is fermentation, and names 9-hydroxycorynantheidine among its products.

Claim 8 above states the biosynthetic order:

corynantheidine
-- (C9 hydroxylase) --> 9-hydroxycorynantheidine
-- (O-methyltransferase) --> mitragynine

9-hydroxycorynantheidine is therefore an intermediate on the route to mitragynine. It is not a compound a manufacturer would have to set out to make. Any facility producing mitragynine by this enzymatic route passes through it and can isolate it at that step.

Both routes also remove the botanical supply chain entirely. No cultivation, no import, no leaf to sample at the border, no alkaloid ratio set by growing conditions. This matters to the Department's scoping because the practical constraint on this compound has been assumed to be its scarcity in the plant. That assumption does not survive either filing.

We note plainly that neither company states any intent to market this compound, that both filings describe general platforms for producing kratom alkaloids with mitragynine as the named target, and that most patent applications are never commercialised. We cite them as evidence that manufacture is feasible and has been claimed, not as evidence of intent.

REQUESTED ACTIONS

1. State whether the Department's scientific and medical evaluation considered mu-opioid-active alkaloids of *Mitragyna speciosa* outside the mitragynine scaffold, including 9-hydroxycorynantheidine, and if so, on what basis they were excluded.
2. State whether the reasoning applied to MGM-16, that a structurally related potent opioid not presently on the market may nonetheless warrant control because of substitution risk, was considered with respect to 9-hydroxycorynantheidine, and if not, why the two were treated differently.
3. State whether the Department is aware of Florida's inclusion of 9-hydroxycorynantheidine in its June 2026 emergency rule, and whether it has reviewed the basis on which that state acted.

4. Publish, in any subsequent notice, the full list of *Mitragyna speciosa* alkaloids considered and the disposition of each, so that the public can evaluate the scope of the action rather than infer it.

5. State whether the evaluation considered that alkaloids scarce in the plant can now be produced enzymatically or by fermentation, and whether the scope of the action was set on natural abundance rather than on manufacturability.

WHAT WE DO NOT ASSERT

We have not located 9-hydroxycorynantheidine products offered for sale, and we do not claim a market presently exists. We do not claim its potency is equivalent to 7-hydroxymitragynine; reported mu-opioid affinity is weaker. We raise a gap in scope, not a finding of present harm.

On the patent applications cited in section 6: both are legitimate biotechnology filings by companies with no connection we have found to any party in the 7-OH proceeding, to any kratom trade association, or to any federal official. Neither states an intent to market 9-hydroxycorynantheidine. We cite them solely as evidence that manufacture at scale is feasible and has been claimed.

Obsidian Watch Group is an independent investigative organization with no financial interest in kratom, 7-OH, or any company in this sector, and no funding from any party to this proceeding.

Respectfully submitted,

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Supporting documents: github.com/jpuckett11/ObsidianGroup_CaseSevenfold